

5 – Post Prototype Refinement of Design

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5.1 Feedback

5.1.1 Approach to Feedback

I gathered feedback in two main ways. Firstly, I created a questionnaire along with a small walkthrough of the system for other peers in my class to fill in. They followed the walkthrough and then answered some prewritten questions on the sheet that I created. This was done without my supervision so I couldn't ask any follow up questions about their feedback. I also did a small presentation to a few of my peers. I demonstrated the system to them and then they verbally gave me feedback. I also asked some questions verbally about what they thought of particular areas.

Name of Respondent	Relationship	Level of Competency
Druvi Patel	Peer	Has similar Java skills to me and is undertaking a similar project.
Daniel Antrobus	Peer	Has similar Java skills to me and is undertaking a similar project.
Harris Beirne	Peer	Has similar Java skills to me and is undertaking a similar project.
James Bridge	Peer	Has similar Java skills to me and is undertaking a similar project.
Nick Wilkie	Tutor	Extensive Java knowledge and experience in creating similar systems.
Josh Stather	Acquaintance	Professional Software developer. Has a large amount of experience in programming similar languages (C#) and has real world experience in creating similar systems.

5.12 Consideration of Feedback

Subject	Feedback	Consideration	Accept Reject	Implications for Redesign
User - View users table	The user table is cramped and should be made larger. The sessions attended can't be seen completely for some users.	This is already planned for the system. See the design of the users screen in the design document.		
User - View users table	Easy and clear to see what each aspect is doing.	I will be keeping the layout of the table the same.		Keep the table layout the same.
User - Add/edit users	It works well. It's easy to input the data. The pop-ups that allow the data to be entered are good. It's good that adding and editing users is consistent because a user is edited in the same way that it's added.	Adding and editing a user are key areas of the final system therefore it's good that the reviewer thinks this area works well.		I will not alter this area too much in the final system.
User - Add/edit users	The process of adding a user is very lengthy and tedious as it is a large amount of pop up boxes. Put the entry on a single screen.	The process of adding a user will not be done very often. I will be keeping the old method of adding a user because the other reviewer said that it was a good and easy to use method of adding a user. However, I will be implementing the method of editing users (detailed in the design) of editing a row of the table because I think it's a simpler way of editing the user's details.		
User - Add/edit users	It's unclear what the cancel button does because when you press it the program continues as if you pressed the ok button. Also, add a facility to cancel adding a user.	I agree with the feedback because, in its current state, the cancel button is confusing because it's non-functional. Being able to cancel creating a user is also an important feature.		I will be making the cancel button work correctly in the final system.

User - Add/edit users	Potentially validate the password that is entered.	It's not important for the passwords of the users to be strong. Implementing a password policy would make the system harder to use for the users because it would mean that they would be more likely to forget their passwords. This also means that more time would be wasted during the sessions resetting passwords.		
User - Search users	It's difficult to find how to reset the search. Add search by last name or username.	Steph and the volunteers already know all of the users by first name so allowing them to search by last name and user name will not make it any easier for them to find users in the system. Adding additional search fields to the view users screen would also introduce more clutter and make the screen more confusing to use.		
User - Delete a user	The table doesn't refresh automatically and the system has to be restarted - this is not ideal. Make the table automatically refresh when a user is deleted to remove the user immediately.	It's confusing that the system has to be restarted for the user account to be removed from the table. I will make the table automatically refresh.		I will make the table automatically refresh.
User - General	The interface is too cramped. Have the table take up multiple rows / have a greater height than the other elements so that it can be larger and easier to see.	This is already planned for the system. See the design of the users screen in the design document.		
View forms - Form attributes	The forms don't have titles and this makes it difficult to quickly identify a form in the system. Add titles to forms.	Although the description is meant to serve the purpose of quickly identifying forms in the system, a title for each form would be better. This is because it will mean that each form can be given a very short and easy to read title. This would be to quickly identify it. This is better than putting it in the description and having to read the entire		The form data structure will need to have a title field added. The view forms table will need an additional column. The create forms screen with need a new field so

		description to work out what the form is. Including a title means that the description of each form could be longer also, this is because currently the description is acting as a title rather than a description.		that the admin can enter a title for the form.
View forms - Filters	It's too complicated as after applying some filters multiple results still remain in the table and it's difficult to see that the filter actually worked. Add an indication that the filter actually worked.	I will be adding this because it could confuse the users if they apply a filter and the table doesn't change because all the forms meet the filter.		Adding some sort of indication that the filter was actually performed would make filtering less confusing for the users. This will require a slight modification to the view forms screen.
View forms/Questions - Overall	The layout is clear (the buttons are large enough to see). It's easy to find the questions and forms in the system.	It's very important that the users can find the forms and questions in the system easily.		I will not alter the overall layout of this screen much in the final system.
View forms/Questions - Table	The table is cramped but works well.	I agree that the table is cramped and a had to read.		I will resize the different columns in the table so that it's easier to read and less cramped.
View forms/Questions - Filters	Add a clear filters/sorts button to add all the forms/questions back to the table.	This is already planned for the system see the design of the view questions and forms screens in the design document		
View forms/Questions - Filters	The text on the radio buttons and checkboxes isn't always visible if the window is too small. Make the window have a minimum width so that it can't be resized so that it's too small for all the labels to be read.	It's important that the text of the checkboxes and radio buttons can be read otherwise it would be very difficult to apply a filter because it would be unclear which options are being selected.		I will add a minimum width to the window to ensure that the labels can always be read.
View Questions -	The layout is confusing.	There are more sorts and filters in the view questions screen that can be applied than in the		

Sorts/Filters	Lay it out like the sorts are laid out in the import/export part of the system.	import/export screen. This means that using the layout of the sorts from the import and export screen wouldn't be feasible as there would be no room for the additional sorts.		
View questions	Multiple questions of the same type with different difficulties is confusing. There isn't a way to immediately tell the difference between the questions which have the same type. Add a title to each question that describes what it is.	This would be a good feature because, currently, the method of viewing questions is very confusing. This is because there is no way to immediately tell the difference between 2 questions of the same type. A title would fix this. This would make finding questions in the system much faster and so would also make creating forms easier and faster for Steph.		The question data structure will need an additional field to store the title. The view questions table will also need an additional column. The create questions screen will need to have a text field added to type the title of the form into.
Attempt form	The red border works well	It's good that this feature works well because it's very important that the users can easily see which questions they have incorrectly answered.		I will keep the red border to indicate which questions have failed their validation checks.
Attempt form	The red border is confusing as it's not immediately obvious what it means. Add an explanation of what the border means.	This is unnecessary because Steph would only need to explain to the users what it meant once and then they wouldn't need further training. Adding the feature to the system could be annoying if it told the users every time they attempted the form. Or, if it was in a help feature, it would not be used very often at all.		
Attempt form	It's unclear to the user how they should fix the question to make the red border disappear. Add a pop-up box that gives the user a suggestion of the correct data type for each question that they got wrong.	While this would make filling the forms easier for the users, it's not realistic therefore I will not be including it. The majority of real word forms do not give the user a suggestion of the data type to enter into a field when they get it wrong. If they became reliant on this feature, in the system, they would struggle without it in the real world.		
Attempt form	Closing the form takes too long and could confuse the	The current amount of time that it takes to close a form is unacceptable. The forms should		

	users. Make closing a form faster.	close much faster as to not waste a large amount of time waiting for the forms to close. However, I am discounting this because faster question saving logic (which would speed up closing a form) is already detailed in the design document under each component type's processing routines. This faster logic was not implemented in the prototype and a slower routine was used.		
Attempt form - Progress saving	Once a form has been attempted once successfully, retrying should reset to the form to empty (perhaps a column on the grid called "Completed" with a counter that increments?)	Currently there is no way for a user to reset a form once they have already completed it. This limits the usefulness of the system because it means that a user can't reattempt a form that they've already completed. This means that if the user has attempted all of the forms in the system they have to wait for a new one to be added if they want to attempt another form. Doing the same things over and over is essential to practicing a skill therefore the ability to attempt the same form multiple times will be a good addition to the system. The counter that tracks how many times a user has completed a form would be a useful statistic to see because it allows Steph and the volunteers to see which forms each user has practised the most.		I will need to alter the attempt form routine so that it checks if a form is completed when the form is attempted, and if it is, it resets the form and increments the counter of how many times the user has completed the form. I will also need to add an additional field to the form in progress class to store the number of times the user has completed a form.
Attempt / Create form	There is no immediate difference between attempting and creating a form, this is confusing. Add something to differentiate these two aspects of the system.	This is unnecessary because only the admin will have the ability to both attempt and a form and create a form. They will spent most of their time creating forms and will not be changing between creating and attempting forms very often therefore there is little potential for confusion.		
Edit forms - Selecting a form	This is nice and clear.	This is good because it's important that it's easy for Steph to select a form to edit in the final system.		I will keep the method of selecting a form to edit using a drop down in the final system.

Create/Edit forms - Sorts	The sort buttons should be smaller and above the table headers.	It's already planned to have the sort buttons below the questions table. See the design of the create forms screen in the design document. Personally, I think that having them below the table looks better than having them above the table.		
Create/Edit forms - Move questions	This works well.	It's important that it's easy for the admin to move questions within a form so that forms can be edited quickly.		I will keep the same method for moving questions in the form in the final system.
Create forms	The form is also intimidating because it's cramped. Add headers and separators to the form.	This would have two advantages. Firstly, it would make the forms more realistic as many online forms have multiple distinct sections. Secondly, would also make the forms less cluttered and clearer to fill in. Currently, the forms are very cluttered and confusing.		I will add a way for the admin to add headers and separators to the forms when they are creating them. This will be added to the create form screen.
Create forms	Add the ability to add optional questions to forms as this is more realistic.	Currently, all of the questions in the form are mandatory. This isn't realistic as the majority of online forms have at least a few questions that aren't mandatory. Adding the ability to add optional questions would give Steph more flexibility when creating forms and would also make the forms that are created more realistic.		I will change the form data structure so that it stores additional information about each question (whether they are required or not). I will also have to redesign parts of the form creation screen to accommodate a method to mark questions as optional. I will also alter the question validation routine.
Create forms - Form preview	A question fills the entire view and resizes when another is entered, that doesn't look too good and is a little confusing if you only have a small number of questions. Make the questions have a	I will be adding this feature because it will make the form preview less cluttered and confusing when there are a small number of questions in the form. This is because the questions won't fill the entire height of the preview and so will not be unnaturally stretched.		The configuration of the JPanel that holds the preview of each question in the form will need to be altered so that the questions have a maximum height.

	maximum height so that they don't fully fill the view if there is a small number of questions.			
Create forms - Add questions	You should be able to view the individual questions in the grid, because often the question type isn't enough information alone. Ability to view questions in the table. (Presumably the question panel).	There is already a button which will open a question to be previewed before adding it to the form. Also, the proposed question title feature will further reduce the need for the ability to see the actual question in the table. This is because it will be easier to identify individual questions. There is also little room in the table to show a preview of each question.		
Create questions	The functionality is present but it would make more sense for it to be a journey rather than a separate screen. Have a button in the view Questions tab to make a question, and clicking that would take you through a dialogue journey where first you state the question name, then the label, then the question type, then the type of control the question is, and then validation, then a preview and then a done button which would save and reload the view question page and close the dialogue.	I will be mostly accepting this piece of feedback. I think that turning the question creation process into a series of dialogs could be tiring, and overwhelming, for the user to fill in. I will instead retain the current create questions screen however, rather than the admin being able to create a question in any order, there will be specific sequential steps that they have to go through (like those suggested). (E.g. add a label and then add a control etc.) This will make creating a question a little less overwhelming and easier for the admin to use		I will need to redesign the create questions screen so that the admin can only complete steps sequentially and also not allow the admins to add multiple rows to a single question. This should simplify the question creation process.
Create questions - Add components	Using buttons to add components to the question is a good way of doing it and better than drag and drop.	It's important that the process of creating questions is as easy as possible so that Steph can create a large number of questions in a small amount of time. Because using buttons is better than drag and drop I will use them in my final system.		Keep the buttons.
Create	This element of	I will keep the method of		Keep the same

questions - Enter final details	creating questions makes a lot of sense and could be incorporated into the journey suggested.	entering the final details the same in the final system. This will make this element of creating questions simple and easy to use.		method of entering final details.
Help feature	Clear and gives good information. Add more details to it about what each button does.	In the interview Steph said that a help feature isn't really required and if one was implemented it should only be very basic. This is because the volunteers and herself would be teaching the users how to use the system anyway and they could just ask the volunteers for help. Therefore, I will not be including this feature.		
Main Menu (lack of)	A main menu is needed. There is no central location of the system. Add a main menu that can serve as a central location in the system that allows access to the other areas.	I agree with this feedback and a main menu was already planned, it was just not included in the prototype. See the design of the main menu in the design document.		

5.2 Input and Output

5.2.1 Consideration

These redesigned screens detail the changes that are to be made to the screens of the application.

Input/Output Facility	New / Existing?	Actions Needed
View forms	Existing	Add an indication that the filter worked. Set a minimum width of the window so that the labels can always be seen. Add additional columns to the table to accommodate the new information.
View questions	Existing	Set a minimum width of the window so that the labels can always be seen. Add additional columns to the table to accommodate the new information.
Create/Edit forms	Existing	Add a button that allows each question to be marked as optional or not. Add a field that allows the user to enter the title of the form.
Add header dialog	New	This needs to be designed and should allow the admin to specify the text they want a header to have when adding it to the form.
Attempt Form	New	An indication of whether each question is optional or not needs to be added.
Create Questions	Existing	This needs to be altered to make creating questions into a journey. It should be altered so that the admin has to create the question in a sequential order. The ability to add multiple rows to a question should also be removed.
Import/Export	Existing	The tables need to be altered to accommodate the additional information about the forms and questions.
Statistics	Existing	An additional column needs to be added to the view questions table to accommodate the additional information stored about each question.

5.22 Changes to Input/Output

View forms

Reconsidered and Restructured: View forms screen

Screens

The screenshot shows a web application titled "Form filler" with a navigation bar containing "Main menu", "Questions", "Forms", "Statistics", and "Import/Export". A table lists two forms:

Type

Re-Design Was Created In DRAW.IO And Is Not An Implementation

Consideration

There is now a small grey filter icon next to the table. This appears when a filter has been applied and disappears when the filter is cleared. (It is visible just to the right of the percentage complete column heading)

In order to ensure that the labels of the check boxes for the main skills tested filter can always be read, the window now has a minimum width. This means that the user can't resize the window to be so narrow that the labels cannot be read.

The title and time completed columns have been added to display their respective data from the form.

The font size of all the text in the table has been reduced to around size 13 (slightly different depending on each column, main skills tested, difficulty, percentage complete, and times completed are size 12). The text has been made smaller to accommodate for the two additional columns in the table (title, and time completed)

The sorts and filters area has also been reduced in width so that there is room for the extra columns in the table.

View Questions

Reconsidered and Restructured: View Questions screen

Screens

Form filler

Main menu Questions Forms Statistics Import/Export

Title	Type	Difficulty
Contact info	Email	4
Birthday	Date picker	6
Street name	Address	2
Gender	Dropdown	4
Photo id	Photo upload	7
Name	Text	2
City	Address	4
Places visited	Check-boxes	2

Sort and Filter

Sorts

Difficulty sort

Type sort

Difficulty filter

1 2 3 4 5 6 7 8 9 10

Apply difficulty filter

Type filter

☒ Email ☐ Birthday ☐ Address

☐ Dropdown ☐ Photo upload ☐ Name

Apply type filter

Reset sorts and filters

Attempt question

Type

Re-Design Was Created In DRAW.IO And Is Not An Implementation

Consideration

In order to ensure that the labels of the radio buttons for the type filter can always be read, the window now has a minimum width. This means that the user can't resize the window to be so narrow that the labels cannot be read.

An additional column has been added to the table so that the title of each question can be displayed. (Title is a new field)

Create/Edit forms

Reconsidered and Restructured: Create/Edit forms screen

Screens

Form filler

Main menu Questions Forms Statistics Create Questions Create Forms Users Import/Export ?

Edit forms

Select a form to edit: Creating new form: F04991976

Form Information

Form title:

Form difficulty: 1

Form description:

Select questions

Title	Type	Difficulty
Basic Email	Email	4
Birthday	Dropdown	6
Street and postcode	Address	2
Country of birth	Dropdown	4

Form Preview

Basic information

Name:

Gender: ☐ Male ☐ Female ☐ other

Type

Re-Design Was Created In DRAW.IO And Is Not An Implementation

Consideration

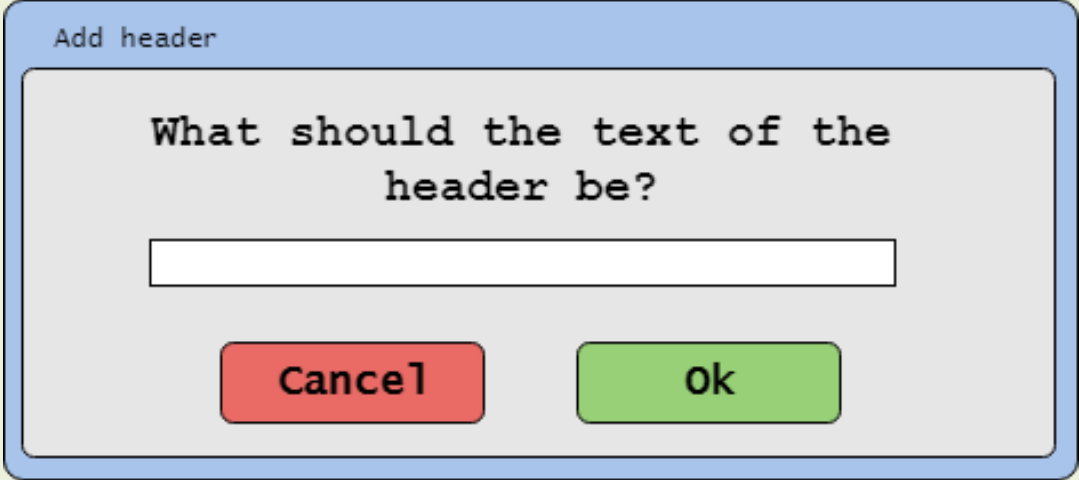
In order to allow a question to be marked as optional, an additional button has been added to the right of each question in the form preview. This button displays a red asterisk when the question is mandatory and is blank when the question is not (in the screenshot the top question is mandatory and the bottom one is not). This is in an attempt to mimic how questions are marked as mandatory in actual online forms (as a red asterisk is used). The button operates as a toggle and by default displays the red asterisk. The remove question button has been changed from a red cross to a rubbish bin icon this is to avoid confusing it with the mark as mandatory button.

There is an additional text box in the form information section which allows the user to enter the title of the form.

There is now an add header button above the reset form button. This opens a dialog which allows the admin to specify the text that the header should have. There is an example header "Basic information" shown in the form preview. There is no option to mark the header as a mandatory question as that would be confusing as the header isn't something that the user should attempt.

An additional column has been added to the select questions table to display each question's title. (This is a new field that has been added to the question data type)

Reconsidered and Restructured: Add header dialog

Screens	
Type	<i>Re-Design Was Created In DRAW.IO And Is Not An Implementation</i>
Consideration	This is a new dialog that I have designed. This dialog will allow the admin to enter the text that a header should have. It will open when they press the add header button in the form creation screen. When they press ok, a header with the specified text will be added to the form.

Attempt form

Reconsidered and Restructured: Attempt form window

Screens	 The screenshot shows a web form titled "Attempt form" with a progress indicator "2 of 4" in the top right corner. The form contains the following fields: Name: A text input field with a red asterisk (*) indicating it is required. Gender: Three radio button options: "Male", "Female", and "Other". The "Other" option is also marked with a red asterisk (*). Password: Two stacked text input fields, both marked with a red asterisk (*). Country: A dropdown menu currently showing "United kingdom", marked with a red asterisk (*). Address line 1: A text input field. Phone number: A text input field. At the bottom of the form are two blue buttons: "Previous" on the left and "Next" on the right.
Type	<i>Re-Design Was Created In DRAW.IO And Is Not An Implementation</i>
Consideration	Each question that is a required question and not optional is marked with a red asterisk. This closely mimics how real online forms show that a question is mandatory.

Create Questions

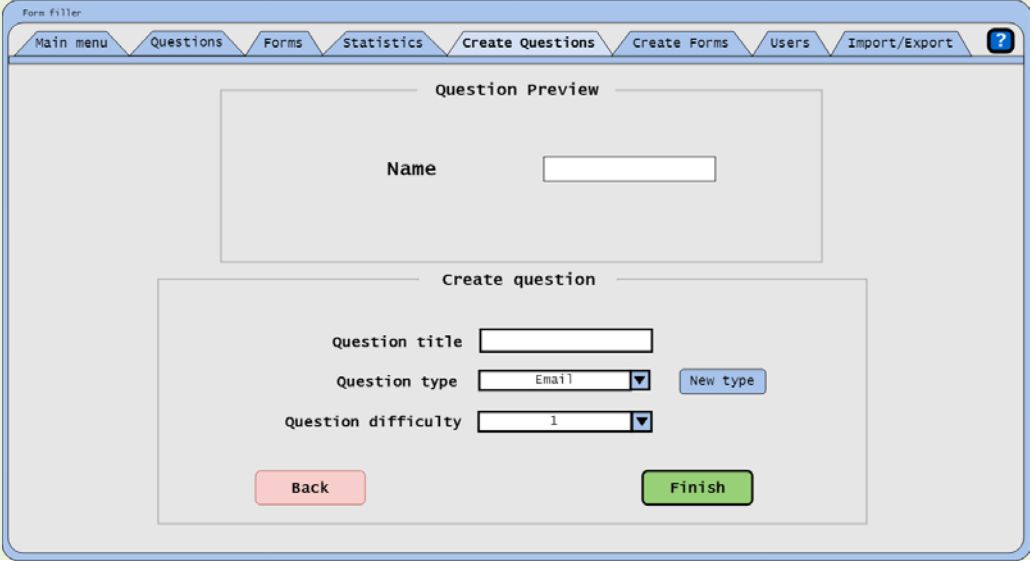
Reconsidered and Restructured: Create questions

Screens

The image displays two sequential screenshots of a web application titled "Form filler". Both screenshots feature a top navigation bar with the following tabs: "Main menu", "Questions", "Forms", "Statistics", "Create Questions" (which is highlighted), "Create Forms", "Users", and "Import/Export". A help icon (?) is located on the far right of the navigation bar.

Top Screenshot: The main content area is divided into two sections. The top section, titled "Question Preview", contains a large, empty rectangular box. The bottom section, titled "Create question", contains a label "Label text:" followed by a text input field. A blue "Next" button is positioned at the bottom right of this section.

Bottom Screenshot: This screenshot shows the next step in the process. The "Question Preview" section now displays the text "Name". The "Create question" section has been updated to include an "Add Component" sub-section. This sub-section contains seven buttons: "Text Field", "Password Field", "Radio Buttons", "Checkboxes", "Drop-down", "Calendar", and "File upload". Below these buttons are two more buttons: a red "Back" button on the left and a blue "Next" button on the right.

	
Type	<i>Re-Design Was Created In DRAW.IO And Is Not An Implementation</i>
Consideration	The question process has been stripped down in light of Josh's feedback. The admin can now only create questions with one row and the creation process is more of a journey. The process starts by the admin entering the text for the label. Then they press next. Then they can add an entry component to the question. (The dialogs that appear when adding a component are not altered from the original design). Once the admin has added a component the buttons are disabled to prevent them from adding multiple components. However, if they press back and then next again they will be able to re-add the component (this is like an undo and is useful if they make a mistake). They can then press next and then enter the final details of the question and press finish. Once they have pressed finish, the question will be saved and the question creation screen will reset and go back to the start. This sequential process of creating a question aims to be less confusing than the previous design. It also makes the role of questions and forms in the system very clear to the admin. The admin could have easily created an entire form in a single question with the previous design. This new design prevents that.

Reconsidered and Restructured: Import/Export

Screens

The screenshot shows the 'Form filler' application interface. At the top, there's a navigation bar with tabs: 'Main menu', 'Questions', 'Forms', 'Statistics', and 'Import/Export'. The 'Import/Export' tab is selected. Below the navigation bar, the screen is split into two main sections: 'Questions' and 'Forms'.

Questions Section:

- At the top is a green 'Import' button.
- Below it is a table with the following data:

Title	Type	Difficulty
Basic Email	Email	4
Birthday	Dropdown	6
Street and postcode	Address	2
Country of birth	Dropdown	4
- Below the table is a large empty rectangular area.
- At the bottom are three buttons: 'Type sort', 'Difficulty sort', and 'Preview'.
- At the very bottom is a blue 'Export' button.

Forms Section:

- At the top is a green 'Import' button.
- Below it is a table with the following data:

Title	Description	Main skills tested	Difficulty
About me	Filling in basic personal info	Name, Dropdown, Birthday, Radiobuttons, Email Address	4
Blue badge practice	Blue badge practice - easy	Name, Dropdown, Address, Radiobuttons, Email Address, Document upload, Date entry	6
- Below the table is a large empty rectangular area.
- At the bottom are two buttons: 'Difficulty sort' and 'Preview'.
- At the very bottom is a blue 'Export' button.

Type

Re-Design Was Created In DRAW.IO And Is Not An Implementation

Consideration

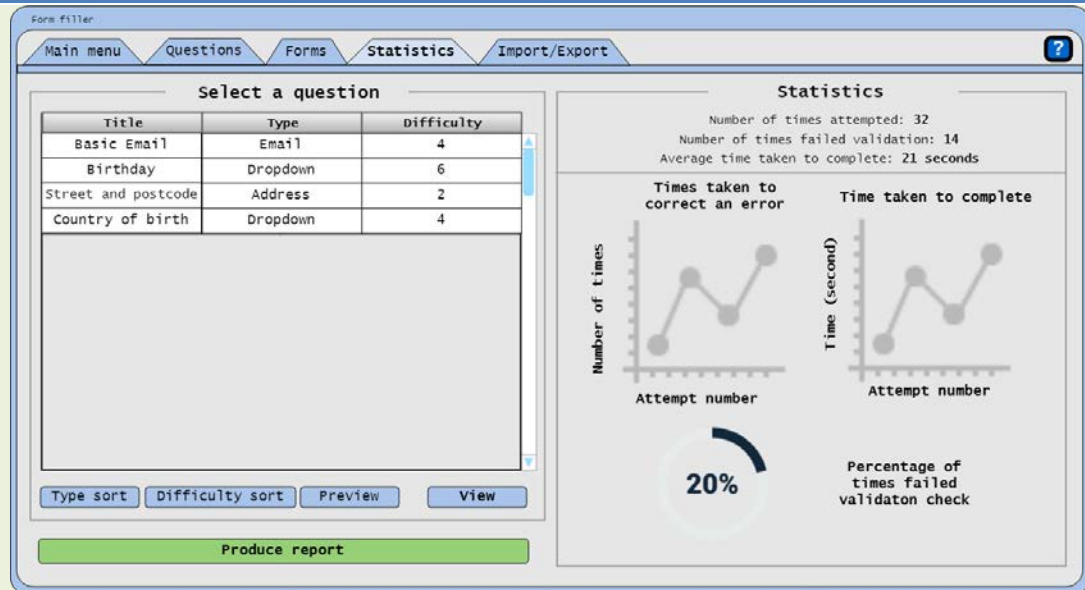
An additional column has been added to the questions table to display each question's title (this is a new field added to the question data type)

An additional column has been added to the forms table to display each form's title (this is a new field added to the form data type)

Statistics

Reconsidered and Restructured: View Statistics screen

Screens



Type

Re-Design Was Created In DRAW.IO And Is Not An Implementation

Consideration

Although it looks like a lot has changed this is just because the statistics screen that was designed in the design wasn't implemented in the prototype. The change that has been made there is that the table with the questions in now has an additional column ("Title") to display the title of each question. Title is a newly added attribute.

5.3 Data Structures & Files

5.31 Consideration

These redesigned data structures detail the changes that are to be made to the data structures of the application.

Data Structure	New / Existing?	Consideration for Additional Data
Form	Existing	The question array needs to be altered so that it can store the header data. An additional array is also needed to store whether each question is required or not. Also a new title field is needed to store the title of each form.
Question	Existing	A new title field is needed to store the title of the question.
FormInProgress	Existing	A new field "timesCompleted" is needed to store the number of times that a user has completed a form.

5.32 Changes to Data Structures

Form - Design

Consideration

The additional array is required so that the system can store which questions in a form are mandatory.

The title field is needed to store the title of the form.

The questions field has been changed to a more general components field. This is to indicate that not only question ids will be stored in it but also the text of the headers.

Field Name	Key Field	Data Type	Validation	Length	Example Data
id	PK	String	-	9	F86189391
components	FK	String[]	-	2-10+ Each String has length 9	Basic details, Q72529010,Q35477436,Address,Q51308304,Q81533244
requiredQuestions	-	Boolean[]	-	2-10+	True,False,True,True
title	-	String	-	10-30	Blue badge form
description	-	String	-	30+	Example blue badge form - easy
mainSkillsTested	-	String[]	-	Size varies with number of question in the Form. Each type is of length 3-10+	Name,Drop-down,Birthday,Address
difficulty	-	int		2	4

Question - Design

Consideration

The title field is needed to store the title of the question.

Field Name	Key Field	Data Type	Validation	Length	Example Data
id	PK	String	-	9	Q72529010
difficulty	-	int	Range check (1-10)	2	3
type	-	String	-	3-10+	Drop-down
title	-	String	-	10-30	Photo id upload

FormInProgress - Design

Consideration

The new times completed field is needed to store the number of times that the user has completed the form.

Field Name	Key Field	Data Type	Validation	Length	Example Data
formID	FK	String	-	9	F86189391
percentComplete	-	int	-	3	25
questionPanels	-	questionPanel[]	-	2-10+	[<Object of type QuestionPanel>, <Object of type QuestionPanel>, <Object of type QuestionPanel>, <Object of type QuestionPanel>]
timesCompleted	-	int	-	2	3

5.4 Processes

5.41 Consideration

These revised and additional processes are required to improve / add new functionality to the system as detailed above in the feedback section.

Entity	Process	New / Existing?	Consideration
FormInProgress List	addCompletedAttempt	New	This method allows the number of times that the form has been completed to be incremented. This method will be called when the user successfully completes a form.
FormInProgress	resetForm	New	This is a routine that takes a form in progress object and resets all of the question panels in it to default i.e. the not filled in state. This is to allow a form to be reset and for the user to be able to complete it a second time
Register	getRegisterOrder	New	This new routine determines which users were present during at least one of the last sessions in the 2 past weeks. This is to make taking the register faster as it means that the system displays the users that it's fairly certain will turn up right at the top of the list. The admin will be able to mark users as present that are not on the register by scrolling further down the list where the rest of the users will be displayed.
Form	Create/Edit forms	Revised	This processing routine is altered to accommodate the new array that stores which questions in the form are mandatory. It has also been altered to accommodate the view title field. The ability to add headers has also been added.
Form	validateForm	Revised	This alteration to the validation logic in the form attempter is required to ensure that questions that are not mandatory only get validated if the user has attempted them. A call to the addCompletedAttempt method of the form in progress object has

			been added. This is to increment the number of times that the user has completed the form once the user successfully completes the form.
Question	Add/Edit	Revised	This alteration to the create questions routine allows the admin to give the question a title.
User	Add/Edit	Revised	This revision to the add/edit user processes is needed to make the cancel button functional.
User	deleteUser	Revised	An additional method call has been added to make the user table automatically refresh when a user is deleted.

5.42 Changes to Processes

// Objective 43 How much of each form a user has attempted should be tracked.

```
START addCompletedAttempt(Parameter: formID){method of FormInProgressList object}

    GLOBAL VARIABLE OF TYPE USER currentLoggedInUser IS PREDECLARED
    GLOBAL VARIABLE OF TYPE FORMINPROGRESSLIST formsInProgress IS PREDECLARED
    DECLARE form IS FORMINPROGRESS

    SET form = formsInProgress.getForm(formID) {getForm is predeclared and gets the
    formInProgress object that corresponds with the formID}

    SET form.timesCompleted = form.timesCompleted + 1

END
```

// NEW: Reset form

```
START resetForm(Parameter: formInProgress) {Takes a formInProgress object and resets all the
question panels}
    QUESTIONPANEL[] ATTRIBUTE questions IS PREDECLARED IN FORMINPROGRESS class
    QUESTIONPANEL ATTRIBUTE id IS PREDECLARED IN QUESTIONPANEL class

    GLOBAL ARRAY OF TYPE QUESTIONPANEL questionPanelArray IS PREDECLARED {has
    been filled with questionPanels from file}
    GLOBAL VARIABLE OF TYPE INTEGER nextQuestionPanelLocation IS PREDECLARED {Next
    free location in the array}

    DECLARE i IS INTEGER
    DECLARE j IS INTEGER

    DECLARE questionPanels IS QUESTIONPANEL[]
    DECLARE emptyPanel IS QUESTIONPANEL
    DECLARE id IS STRING

    SET questionPanels = formInProgress.questions

    FOR i=0 TO questionPanels.length DO
        SET id = questionPanels[i].id
        FOR j=0 TO nextQuestionPanelLocation DO
            IF questionPanelArray[j].id == id THEN
                SET emptyPanel = questionPanelArray[j]
                BREAK
            ENDIF
        ENDFOR

        SET questionPanels[i] = emptyPanel
    ENDFOR

    SET formInProgress.questions = questionPanels {Overwrite the filled in panel with the
    gathered empty ones}

END
```

```

// NEW: Objective 63 The admin user should be able to take a register

START getRegisterOrder()
  GLOBAL ARRAY OF TYPE USER[] userArray IS PREDECLARED {has been filled with users
    from file}
  GLOBAL VARIABLE OF TYPE INTEGER nextUserLocation IS PREDECLARED {Next free
    location in the array}

  STRING ARRAY ATTRIBUTE sessionsPresentAt is PREDECLARED IN USER class

  DECLARE topUsersInRegister IS USER[nextUserLocation]
  DECLARE restOfUsersInRegister IS USER[nextUserLocation]
  DECLARE nextRegisterLocation IS INTEGER
  DECLARE nextRestOfRegisterLocation IS INTEGER

  DECLARE mostRecentSession IS STRING
  DECLARE secondMostRecentSession IS STRING

  DECLARE sessionsAt IS STRING[]
  DECLARE mostRecentSessionAttended IS STRING

  DECLARE i IS INTEGER

  SET mostRecentSession = getMostRecentSession() {Predeclared and returns the date of the
    most recent session}
  SET secondMostRecentSession = getSecondMostRecentSession() {Predeclared and returns
    the date of the second most recent session}

  SET nextRegisterLocation = 0
  SET nextRestOfRegisterLocation = 0

  FOR i=0 TO i=nextUserLocation DO {Search through the list of users to find which were
    present}
    SET sessionsAt = userArray[i].sessionsPresentAt
    SET mostRecentSessionAttended = sessionsAt[sessionsAt.length-1] {Get the last
      item in the array}

    IF mostRecentSessionAttended == mostRecentSession OR
      mostRecentSessionAttended == secondMostRecentSession THEN
      SET topUsersInRegister[nextRegisterLocation] = userArray[i]
      SET nextRegisterLocation = nextRegisterLocation + 1
    ELSE
      SET restOfUsersInRegister[nextRestOfRegisterLocation] = userArray[i]
      SET nextRestOfRegisterLocation = nextRestOfRegisterLocation + 1
    ENDIF
  ENDFOR

  {Both arrays are made now}

  DECLARE finalUserList IS USER[nextUserLocation]

  FOR i=0 to i=nextUserLocation
    IF i >= nextRegisterLocation THEN
      SET finalUserList[i] = restOfUsersInRegister[i-nextRegisterLocation]
    ELSE
      SET finalUserList[i] = topUsersInRegister[i]
    ENDIF
  ENDFOR

```

```

ENDFOR

RETURN finalUserList

END

```

// Revised: Objective 25 Ability to create forms

START

GLOBAL ARRAY OF TYPE FORM formArray IS PREDECLARED {has been filled with forms from file}

GLOBAL VARIABLE OF TYPE INTEGER nextFormLocation IS PREDECLARED {Next free location in the array}

DECLARE newForm IS NEW Form()

STRING ATTRIBUTE id IS PREDECLARED IN FORM class

STRING[] ATTRIBUTE **components** IS PREDECLARED IN FORM class

BOOLEAN[] ATTRIBUTE requiredQuestions IS PREDECLARED IN FORM class

STRING ATTRIBUTE description IS PREDECLARED IN FORM class

STRING ATTRIBUTE title IS PREDECLARED IN FORM class

STRING[] ATTRIBUTE mainSkillsTested IS PREDECLARED IN FORM class

STRING ATTRIBUTE difficulty IS PREDECLARED IN FORM class

DECLARE newId IS STRING

DECLARE more**Components** IS BOOLEAN

DECLARE i IS INTEGER

DECLARE new**Components**[100] IS STRING ARRAY

DECLARE newRequiredQuestions[100] IS BOOLEAN ARRAY

DECLARE newMainSkillsTested[100] IS STRING ARRAY

SET newId = CALL METHOD getFreeID() {getFreeID() is predeclared}

INPUT newTitle

INPUT newDescription

INPUT newDifficulty

SET moreQuestions = TRUE

SET i = 0

WHILE more**Components** DO

INPUT new**Components**[i]

IF newComponents[i] OF TYPE QuestionID THEN

INPUT newRequiredQuestions[i]

SET newMainSkillsTested[i] = newQuestions[i].mainSkillTested

ELSE THEN

SET newRequiredQuestions[i] = FALSE

ENDIF

OUTPUT "Continue entering **components**? "

INPUT response

IF response == "n" THEN {If they entered n (for no)}

SET more**Components** = FALSE

ENDIF

SET i = i + 1

```

ENDWHILE

DECLARE trimmedComponents[i] IS STRING ARRAY
DECLARE trimmedRequiredQuestions[i] IS BOOLEAN ARRAY
DECLARE trimmedSkills[i] IS STRING ARRAY
DECLARE j IS INTEGER
FOR j=0 TO i DO
    trimmedComponents[j] = newComponents[j]
    trimmedRequiredQuestions[j] = newRequiredQuestions[j]
    trimmedSkills[j] = newMainSkillsTested[j]

ENDFOR

SET newForm.id = newId
SET newForm.components = trimmedComponents
SET newForm.requiredQuestions = trimmedRequiredQuestions
SET newForm.title = newTitle
SET newForm.description = newDescription
SET newForm.mainSkillsTested = trimmedSkills
SET newForm.difficulty = newDifficulty

SET formArray[nextFormLocation] = newForm

CALL METHOD writeDatabase()

END

```

// Revised: Objective 30 Ability to edit forms

```

START editUser(Parameter: formToEdit)
    STRING ATTRIBUTE id IS PREDECLARED IN FORM class
    STRING[] ATTRIBUTE components IS PREDECLARED IN FORM class
    BOOLEAN[] ATTRIBUTE requiredQuestions IS PREDECLARED IN FORM class
    STRING ATTRIBUTE description IS PREDECLARED IN FORM class
    STRING ATTRIBUTE title IS PREDECLARED IN FORM class
    STRING[] ATTRIBUTE mainSkillsTested IS PREDECLARED IN FORM class
    STRING ATTRIBUTE difficulty IS PREDECLARED IN FORM class

    DECLARE moreComponents IS BOOLEAN
    DECLARE i AS INTEGER
    DECLARE newComponents [100] IS STRING ARRAY
    DECLARE newRequiredQuestions [100] IS BOOLEAN ARRAY
    DECLARE newMainSkillsTested[100] IS STRING ARRAY

    OUTPUT formToEdit.id
    INPUT response
    IF response != "" THEN {If they entered a new username}
        SET formToEdit.id = response
    ENDIF

    OUTPUT formToEdit.title
    INPUT response
    IF response != "" THEN
        SET formToEdit.title = response
    ENDIF

    OUTPUT formToEdit.description

```

```

INPUT response
IF response != "" THEN
    SET formToEdit.description = response
ENDIF

OUTPUT formToEdit.difficulty
INPUT response
IF response != "" THEN
    SET formToEdit.difficulty = response
ENDIF

OUTPUT formToEdit.components + " enter 'y' to recreate the list"
INPUT response
IF response == 'y' THEN

    SET moreComponents = TRUE
    SET i = 0
    WHILE moreComponents DO
        INPUT newComponents[i]
        IF newComponents[i] OF TYPE QuestionID THEN
            INPUT newRequiredQuestions[i]
            SET newMainSkillsTested[i] = newQuestions[i].mainSkillTested
        ELSE THEN
            SET newRequiredQuestions[i] = FALSE
        ENDIF

        OUTPUT "Continue entering components? "
        INPUT response
        IF response == "n" THEN {If they entered n (for no)}
            SET moreComponents = FALSE
        ENDIF
        SET i = i + 1
    ENDWHILE

    DECLARE trimmedComponents[i] IS STRING ARRAY
    DECLARE trimmedRequiredQuestions[i] IS BOOLEAN ARRAY
    DECLARE trimmedSkills[i] IS STRING ARRAY
    DECLARE j is INTEGER
    FOR j=0 TO i DO
        trimmedComponents[j] = newComponents[j]
        trimmedSkills[j] = newMainSkillsTested[j]
        trimmedRequiredQuestions[j] = newRequiredQuestions[j]
    ENDFOR
    SET formToEdit.components= trimmedComponents
    SET formToEdit.requiredQuestions = trimmedRequiredQuestions
    SET formToEdit.mainSkillsTested = trimmedSkills
ENDIF

CALL METHOD writeDatabase()

```

END

// Revised: Objective 35 Ability to submit a form

```
START validateForm()
  GLOBAL VARIABLE OF TYPE FORM formInProgress IS PREDECLARED {The form that the
    user is currently attempting}
  GLOBAL VARIABLE OF TYPE USER currentLoggedInUser IS PREDECLARED
  GLOBAL VARIABLE OF TYPE FORMINPROGRESSLIST formsInProgress IS PREDECLARED

  GLOBAL VARIABLE OF TYPE QUESTIONPANEL[] questions IS PREDECLARED {Contains the
    questions that are in the form}

  GLOBAL VARIABLE OF TYPE INTEGER[] failedValidationChecks IS PREDECLARED
  GLOBAL VARIABLE OF TYPE LONG[] timeToCompleteQuestions IS PREDECLARED

  DECLARE i IS INTEGER
  DECLARE allCorrect IS BOOLEAN
  DECLARE passed IS BOOLEAN

  FOR i=0 TO questions.length DO
    IF formInProgress.requiredQuestions[i] OR questions[i].presenceCheck() THEN {If
      the question is required, or the question passed a presence check i.e. the user has
      attempted it}
      SET passed = questions[i].validate() {validate() is predeclared in each type
        of question class}

      IF !passed THEN
        SET allCorrect = FALSE
        currentLoggedInUser.addFailedValidation(questions[i])
        {addFailedAttempt is predeclared}
        SET failedValidationChecks[i] = failedValidationChecks[i] + 1
      ENDIF
    ENDIF
  ENDFOR

  IF allCorrect THEN
    FOR i=0 TO questions.length DO
      IF failedVaildationChecks[i] > 0 THEN
        currentLoggedInUser.addNumberOfAttemptsNeededToCorrect(ques
          tions[i], failedValidationChecks[i]) {addNumberOfAttempts is
          predeclared}
      ENDIF
      currentLoggedInUser.addTimeTakenToComplete(questions[i],
        timeToCompleteQuestions[i]) {addTimeTakenToComplete is predeclared}
    ENDFOR

    formsInProgress.addCompletedAttempt(formInProgress.id)

    OUTPUT "Form Complete"
  ENDIF
END
```

// Revised: Objective 1 Ability to create Questions

START createQuestion()

GLOBAL ARRAY OF TYPE QUESTION questionArray IS PREDECLARED {has been filled with questions from file}

GLOBAL VARIABLE OF TYPE INTEGER nextQuestionLocation IS PREDECLARED {Next free location in the array}

DECLARE newQuestion IS NEW Question()

STRING ATTRIBUTE id IS PREDECLARED IN QUESTION class

STRING ATTRIBUTE difficulty IS PREDECLARED IN QUESTION class

STRING ATTRIBUTE type IS PREDECLARED IN QUESTION class

STRING ATTRIBUTE title IS PREDECLARED IN QUESTION class

DECLARE newId IS STRING

SET newId = CALL METHOD getFreeID() {getFreeID() is predeclared}

INPUT newDifficulty

INPUT newType

INPUT newTitle

SET newQuestion.id = newId

SET newQuestion.difficulty = newDifficulty

SET newQuestion.type = newType

SET newQuestion.title = newTitle

createQuestionPanel(newID)

SET questionArray[nextQuestionLocation] = newQuestion

CALL METHOD writeDatabase()

END

// Revised: Objective 38 Ability to add users

START

GLOBAL ARRAY OF TYPE USER[] userArray IS PREDECLARED {has been filled with users from file}

GLOBAL VARIABLE OF TYPE INTEGER nextUserLocation IS PREDECLARED {Next free location in the array}

DECLARE newUser IS NEW User()

STRING ATTRIBUTE id IS PREDECLARED IN USER class

STRING ATTRIBUTE username IS PREDECLARED IN USER class

STRING ATTRIBUTE password IS PREDECLARED IN USER class

STRING ATTRIBUTE firstName IS PREDECLARED IN USER class

STRING ATTRIBUTE lastName IS PREDECLARED IN USER class

STRING ATTRIBUTE dateOfBirth IS PREDECLARED IN USER class

STRING ATTRIBUTE phoneNumber IS PREDECLARED IN USER class

STRING ARRAY ATTRIBUTE sessionsPresentAt is PREDECLARED IN USER class

DECLARE newId IS STRING

DECLARE moreSessions IS BOOLEAN

DECLARE i as INTEGER

DECLARE newSessions[100] IS STRING ARRAY

DECLARE cancelled IS BOOLEAN {Stores whether the user pressed the cancel button}

SET newId = CALL METHOD getFreeID() {getFreeID() is predeclared}

INPUT newUsername

IF cancelled THEN

RETURN {exit the routine}

ENDIF

INPUT newPassword

IF cancelled THEN

RETURN {exit the routine}

ENDIF

INPUT newFirstName

IF cancelled THEN

RETURN {exit the routine}

ENDIF

INPUT newLastName

IF cancelled THEN

RETURN {exit the routine}

ENDIF

INPUT newDateOfBirth

IF cancelled THEN

RETURN {exit the routine}

ENDIF

INPUT newPhoneNumber

IF cancelled THEN

RETURN {exit the routine}

ENDIF

SET moreSessions = TRUE

SET i = 0

WHILE moreSessions DO

INPUT newSessions[i]

IF cancelled THEN

RETURN {exit the routine}

ENDIF

OUTPUT "Continue entering sessions? "

INPUT response

IF response == "n" THEN {If they entered n (for no)}

SET moreSessions = FALSE

ENDIF

SET i = i + 1

ENDWHILE

DECLARE trimmedSessions[i] IS STRING ARRAY

DECLARE j is INTEGER

FOR j=0 TO i DO

trimmedSessions[j] = newSessions[j]

ENDFOR

SET newUser.id = newId

SET newUser.username = newUsername

SET newUser.password = newPassword

SET newUser.firstName = newFirstName

SET newUser.lastName = newLastName

SET newUser.dateOfBirth = newDateOfBirth

SET newUser.phoneNumber = newPhoneNumber

```

SET newUser.sessionsPresentAt = trimmedSessions

SET userArray[nextUserLocation] = newUser

CALL METHOD writeDatabase()

END

```

// Revised: Objective 39 Ability to edit users

```

START editUser(Parameter: userToEdit)
  STRING ATTRIBUTE id IS PREDECLARED IN USER class
  STRING ATTRIBUTE username IS PREDECLARED IN USER class
  STRING ATTRIBUTE password IS PREDECLARED IN USER class
  STRING ATTRIBUTE firstName IS PREDECLARED IN USER class
  STRING ATTRIBUTE lastName IS PREDECLARED IN USER class
  STRING ATTRIBUTE dateOfBirth IS PREDECLARED IN USER class
  STRING ATTRIBUTE phoneNumber IS PREDECLARED IN USER class

  STRING ARRAY ATTRIBUTE sessionsPresentAt is PREDECLARED IN USER class

  DECLARE moreSessions IS BOOLEAN
  DECLARE i as INTEGER
  DECLARE newSessions[100] IS STRING ARRAY

  DECLARE cancelled IS BOOLEAN {Stores whether the user pressed the cancel button}

  OUTPUT userToEdit.username
  INPUT response
  IF cancelled THEN
    RETURN {exit the routine}
  ENDIF
  IF response != "" THEN {If they entered a new username}
    SET userToEdit.username = response
  ENDIF

  OUTPUT userToEdit.password
  INPUT response
  IF cancelled THEN
    RETURN {exit the routine}
  ENDIF
  IF response != "" THEN
    SET userToEdit.password = response
  ENDIF

  OUTPUT userToEdit.firstName
  INPUT response
  IF cancelled THEN
    RETURN {exit the routine}
  ENDIF
  IF response != "" THEN
    SET userToEdit.firstName = response
  ENDIF

  OUTPUT userToEdit.lastName
  INPUT response
  IF cancelled THEN

```

```

        RETURN {exit the routine}
    ENDIF
    IF response != "" THEN
        SET userToEdit.lastName = response
    ENDIF

    OUTPUT userToEdit.dateOfBirth
    INPUT response
    IF cancelled THEN
        RETURN {exit the routine}
    ENDIF
    IF response != "" THEN
        SET userToEdit.dateOfBirth = response
    ENDIF

    OUTPUT userToEdit.phoneNumber
    INPUT response
    IF cancelled THEN
        RETURN {exit the routine}
    ENDIF
    IF response != "" THEN
        SET userToEdit.phoneNumber = response
    ENDIF

    OUTPUT userToEdit.sessionsPresentAt + " enter 'y' to recreate the list"
    INPUT response
    IF cancelled THEN
        RETURN {exit the routine}
    ENDIF
    IF response == 'y' THEN

        SET moreSessions = TRUE
        SET i = 0
        WHILE moreSessions DO
            INPUT newSessions[i]
            IF cancelled THEN
                RETURN {exit the routine}
            ENDIF
            OUTPUT "Continue entering sessions? "
            INPUT response
            IF response == "n" THEN {If they entered n (for no)}
                SET moreSessions = FALSE
            ENDIF
            SET i = i + 1
        ENDWHILE

        DECLARE trimmedSessions[i] IS STRING ARRAY

        DECLARE j is INTEGER
        FOR j=0 TO i DO
            trimmedSessions[j] = newSessions[j]
        ENDFOR

        SET userToEdit.sessionsPresentAt = trimmedSessions
    ENDIF

    CALL METHOD writeDatabase()

```

END

// Revised: Objective 40 Ability to delete users

```
START deleteUser(Parameter: userIDToDelete)
  GLOBAL ARRAY OF TYPE USER userArray IS PREDECLARED {has been filled with users
  from file}
  GLOBAL VARIABLE OF TYPE INTEGER nextUserLocation IS PREDECLARED {Next free
  location in the array}
  DECLARE newArray[userArray.length] IS USER[]
  DECLARE i IS INTEGER
  DECLARE j IS INTEGER

  SET j = 0

  FOR i=0 TO nextUserLocation DO
    IF (userArray[i].ID != userIDToDelete) THEN {If the user isn't the one that is to be
    deleted}
      SET newArray[j] = userArray[i]
      SET j = j + 1
    ENDIF
  ENDFOR

  SET userArray = newArray
  SET nextUserLocation = nextUserLocation - 1

  CALL METHOD writeDatabase()
  CALL METHOD refreshTable() {Predeclared and refreshes the table}
END
```